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LLM Training & Power Fluctuations

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Motivation

Power Swings

- Power swings in training can align with resonant frequencies of grid components
- Documented occurrences by utility providers

Scale

- Fluctuation profile is inherent to communication & checkpointing in pipelined training environments
- Model size requires tens-of-thousands of GPUs, increasing size of fluctuations

Primary Idea

- Systems-level approaches to deal with power

There are also "grid-side" approaches

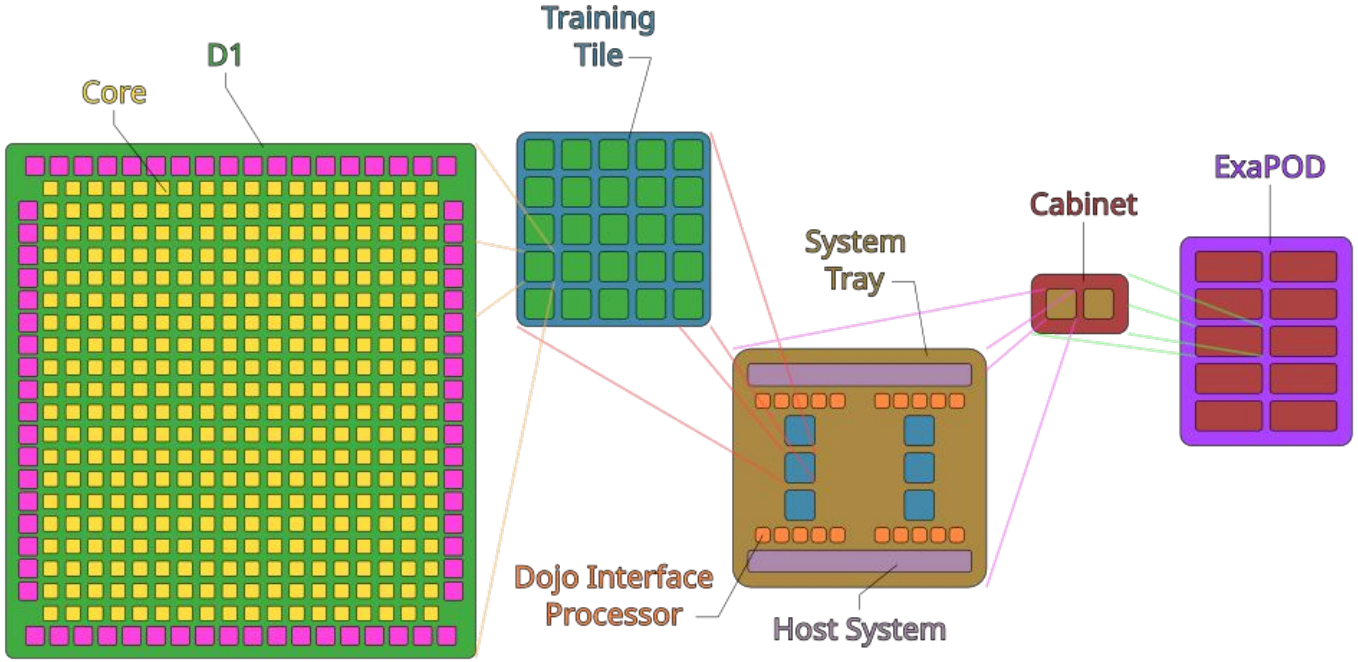
Not talking about specifically (largely due to scale):

- Low-power compute (e.g. waferscale)
- EDP (energy-delay product)
- microarchitectural or architectural approaches

Other Real Apps!

Tesla's Dojo (rip 8/2025)

- Took down whole grids



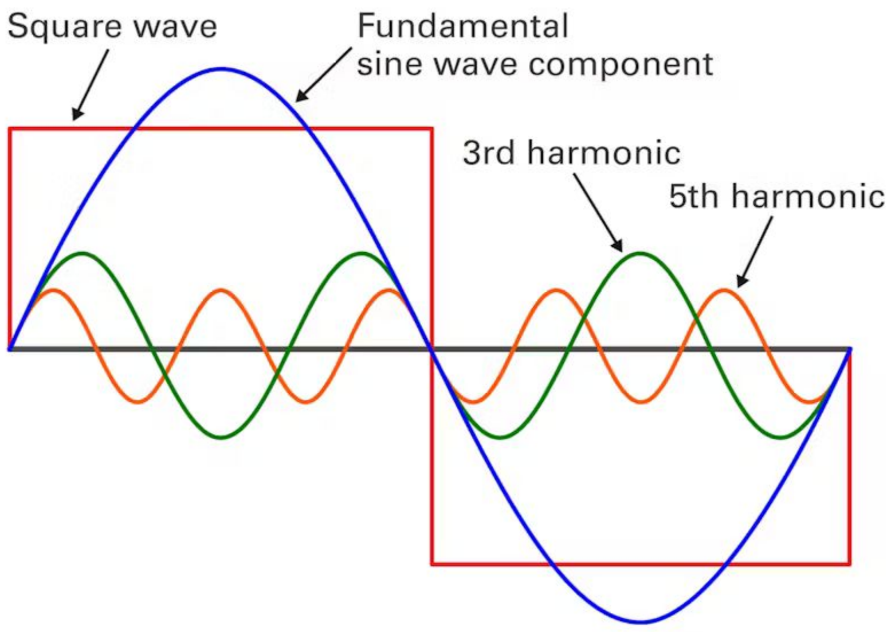
How Do Grids Go Down????

"Fundamental" frequency:

50Hz in most places

60Hz in others (North America)

- Next gen moving from specific freqs



Generally speaking:

Oscillating, non-uniform load

Traditional Prevention

Active and/or passive harmonic filters

- Passive:

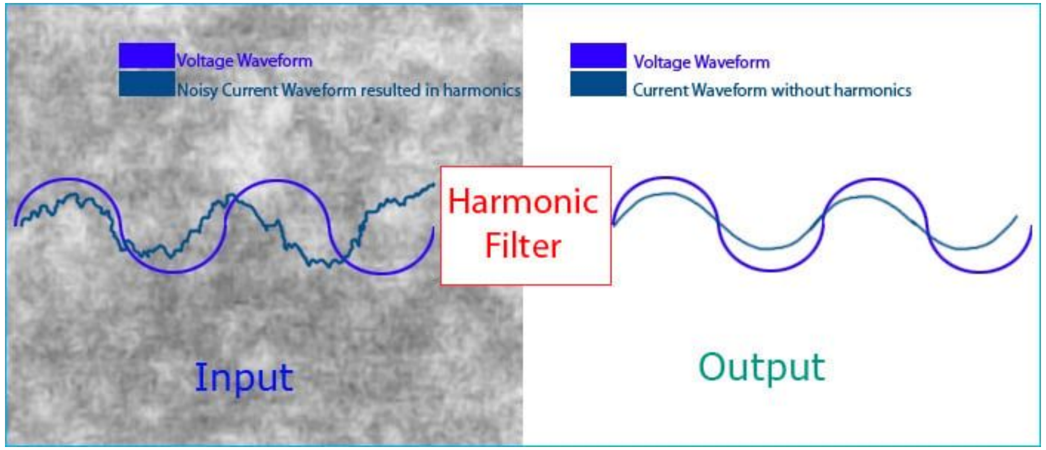
  Divert currents generated by harmonics

  Bad filters → more resonance lol

- Active:

  Monitor and “cancel out” harmonics

  Send matching frequency 180 out of phase



Problem

AI Training Load Fluctuations at Gigawatt-Scale

- Tells us about power grids & infra
- Influence of swings in general
- Load better be stable

Power Consumption Profile

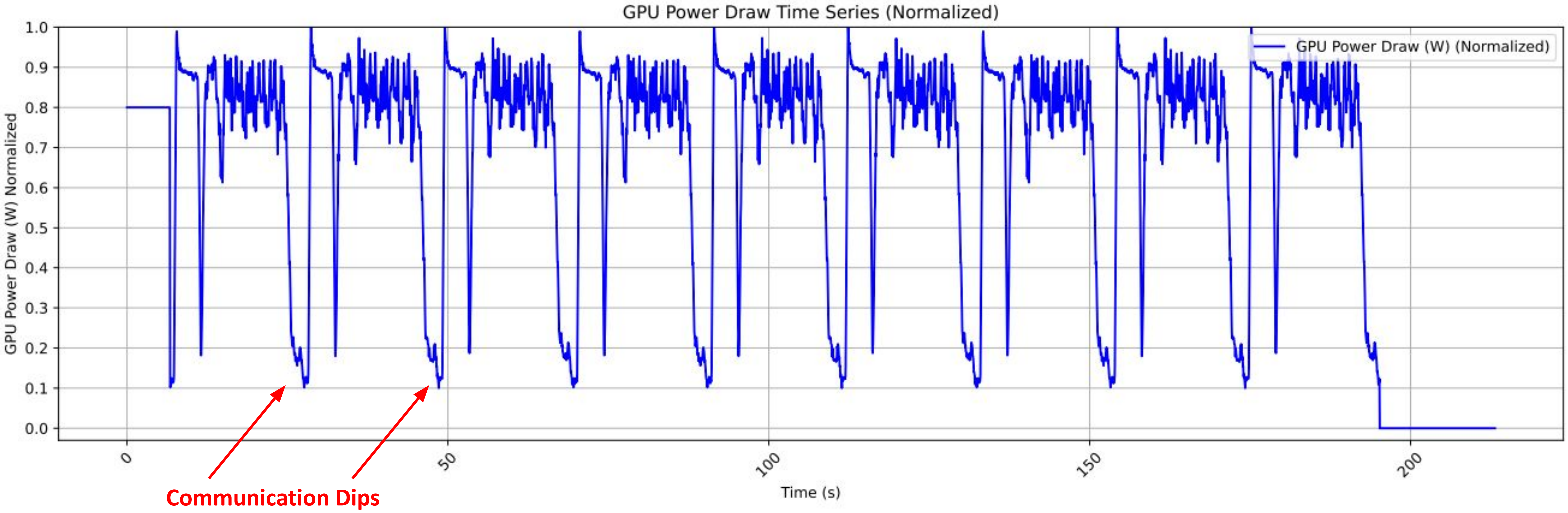
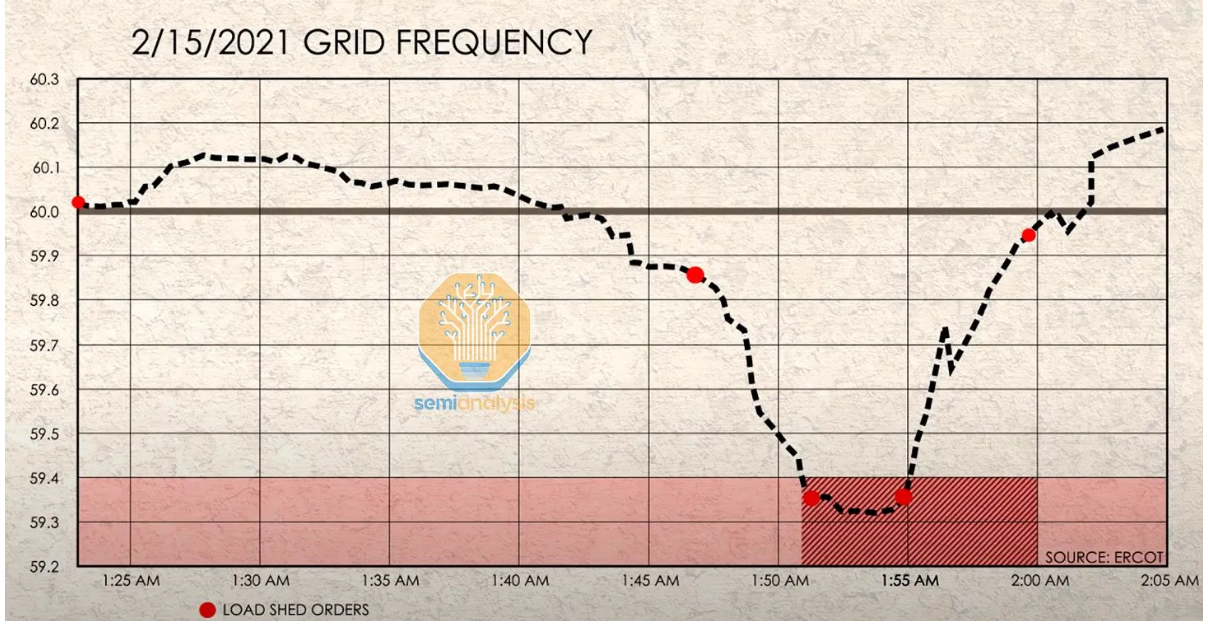


Fig. 1. Power readings from an at-scale training job on DGX-H100 racks.

Stability & Load

Stable/easy to model:

- Household
- Manufacturing
- Chip fabrication
- Trad. datacenters



Unstable/hard to model:

- AI Load ???????

AI Load Fluctuation - Training

Why?

~~order of milliseconds

- spike during compute, dip during comms
- massive dip during checkpointing

~~order of seconds

- Idle GPUs during large syncs (AllReduce)

End of run load drop

AI Load Fluctuation - Inference

Same principles as training

Prefill vs. decode

- Prefill high util
- Decode low util

More communication stalls…

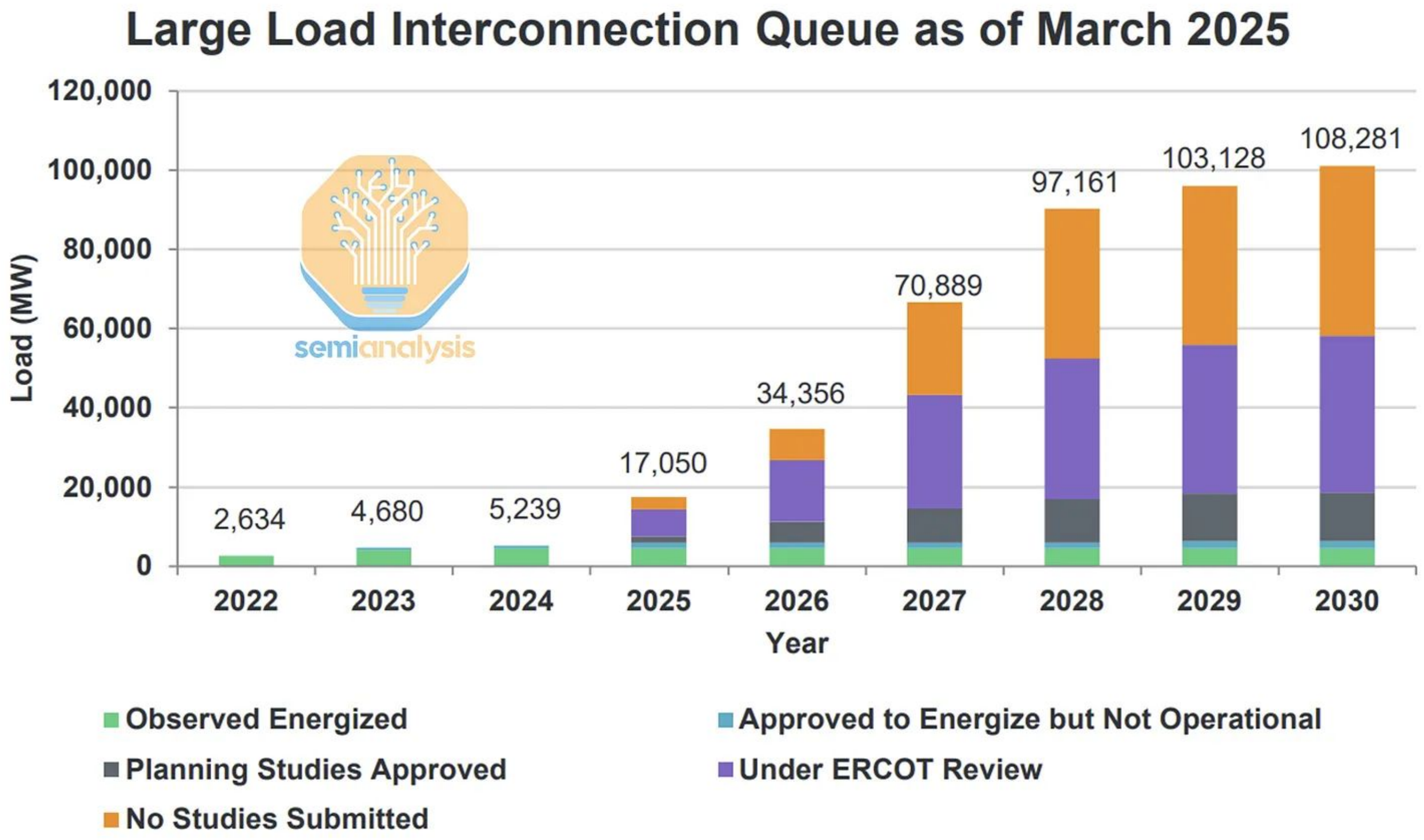
How Worried Should We Be?

ERCOT's estimates:

(reliability council)

Unrealistic, but why?

Notice how few have studies or approval



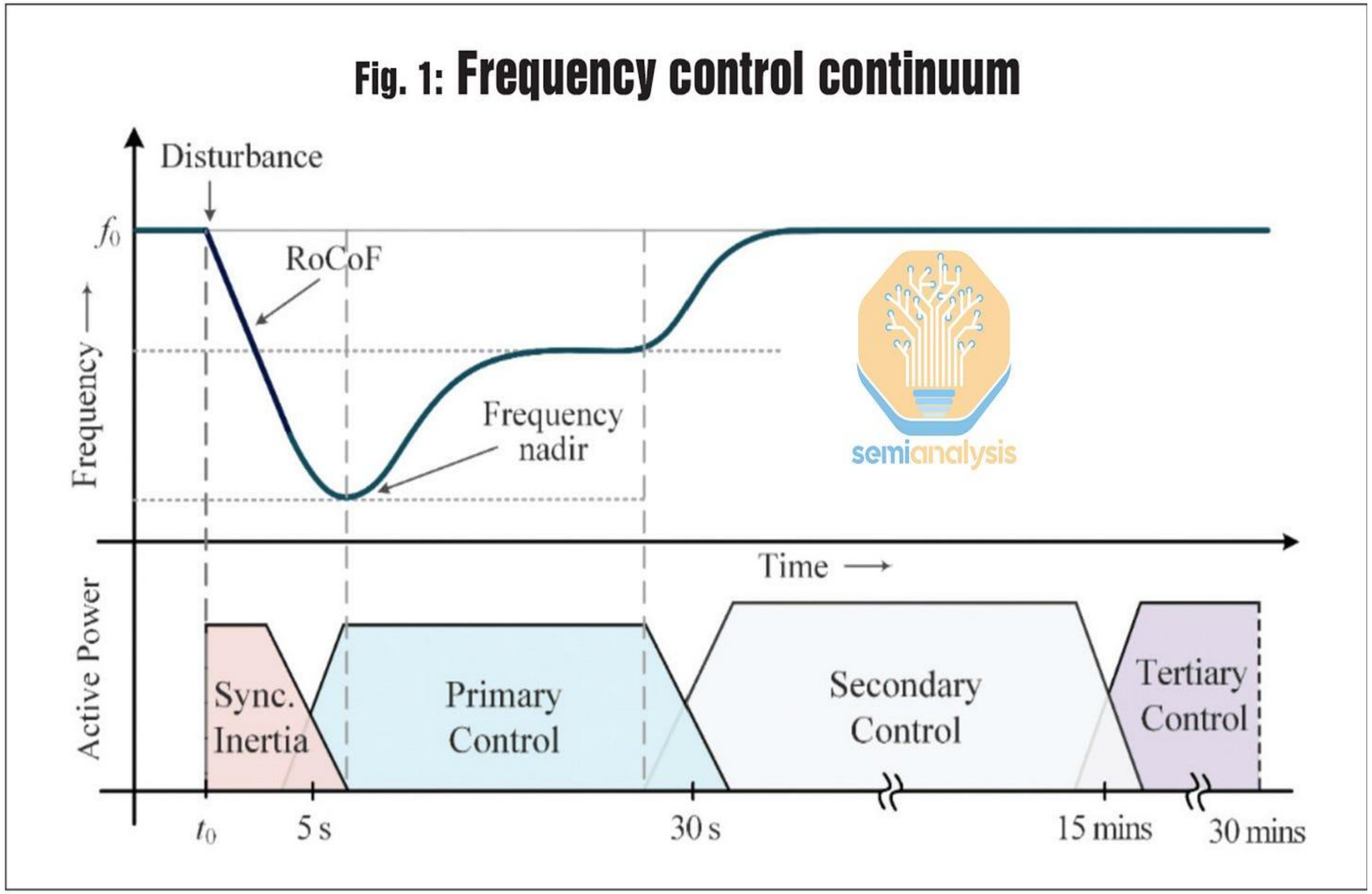
AI Training Load Fluctuations at Gigawatt-scale - Risk of Power Grid Blackout?

Jeremie Eliahou Ontiveros, Dylan Patel, and Ajey Pandey

Problems of Concern

- Managing the Fast Power Fluctuations
- Risk of Cascading Blackouts
- Datacenter Disconnection
- Cascade Risk
- Situation from Iberian Peninsula

Managing Fast Power Fluctuations



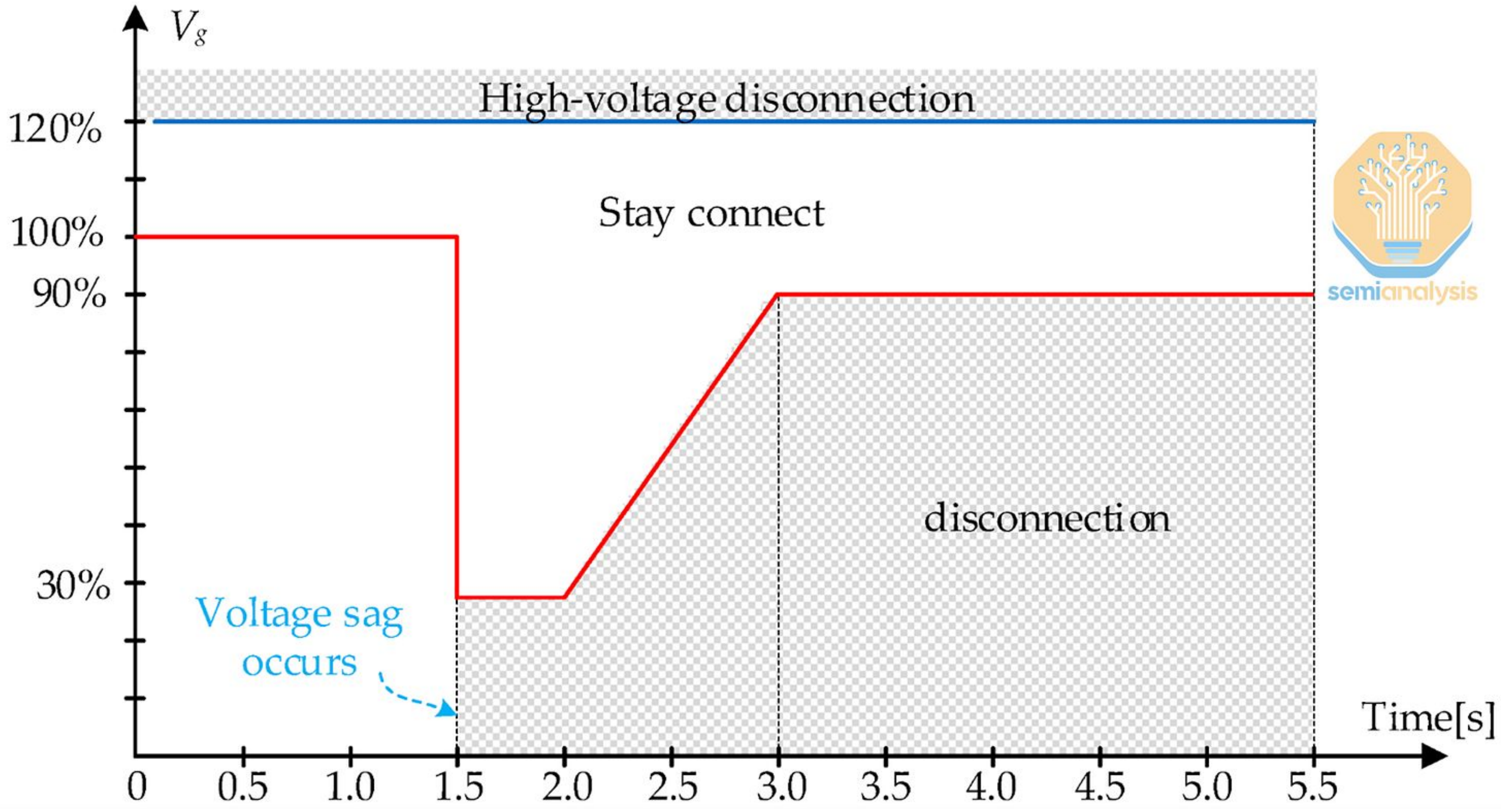
Fluctuations usually handled at MW scale

Generators can ramp output by 100MW within 10 minutes

How does the backup currently work??

Cascading Blackouts

Low voltage ride throughs



Cascading Blackouts

Uninterrupted power supplies (URPS) have to kick in once disconnected

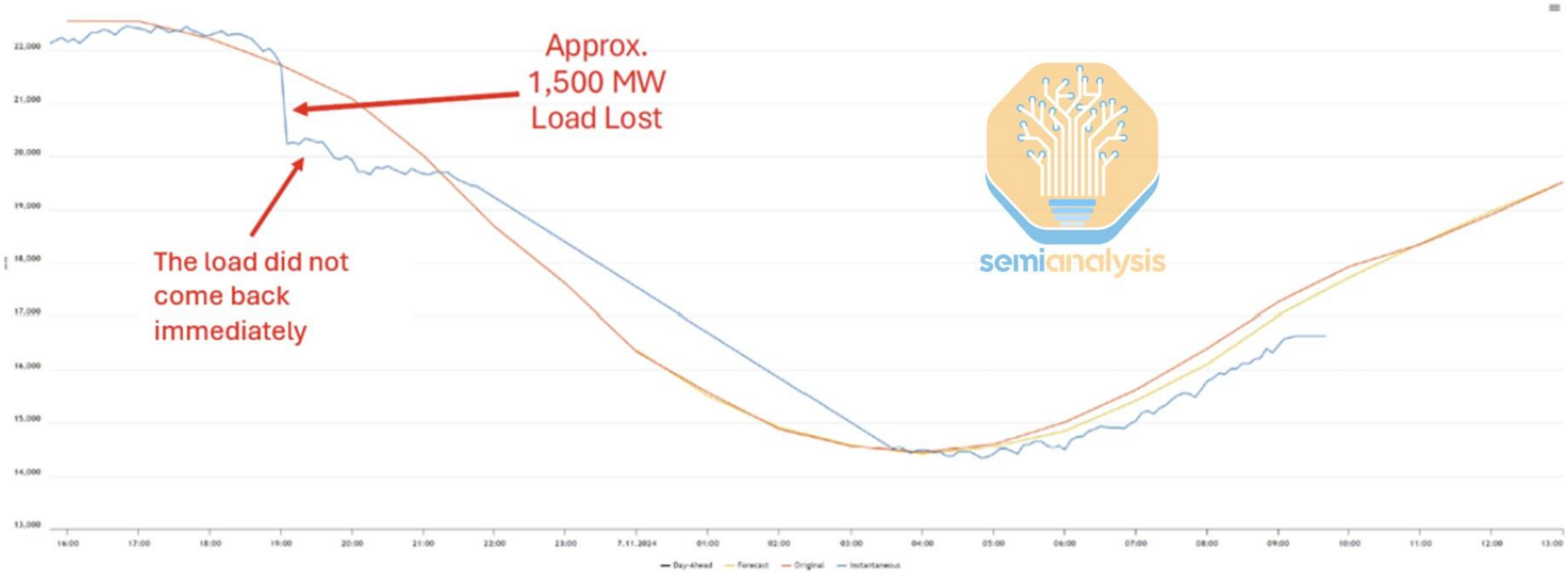


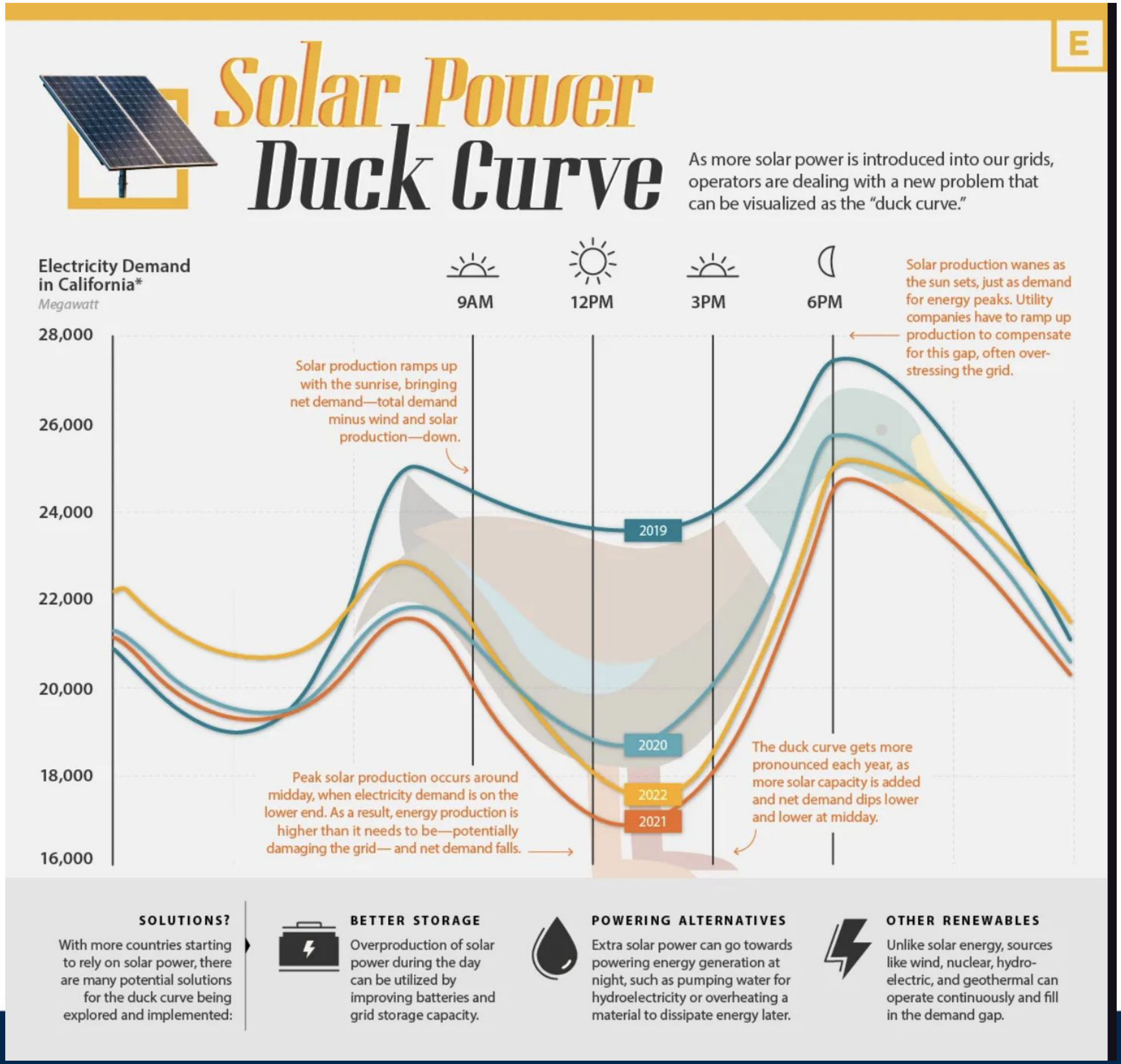
Figure 2: System Load Chart

Datacenter Disconnection

If a typical grid fault happens in West Texas, how many datacenters would disconnect instantly?

Duck curve: Midday is low demand and extremely high solar power with low inertia in the system

Summer pattern (SP): High power demand, peak heat wave season



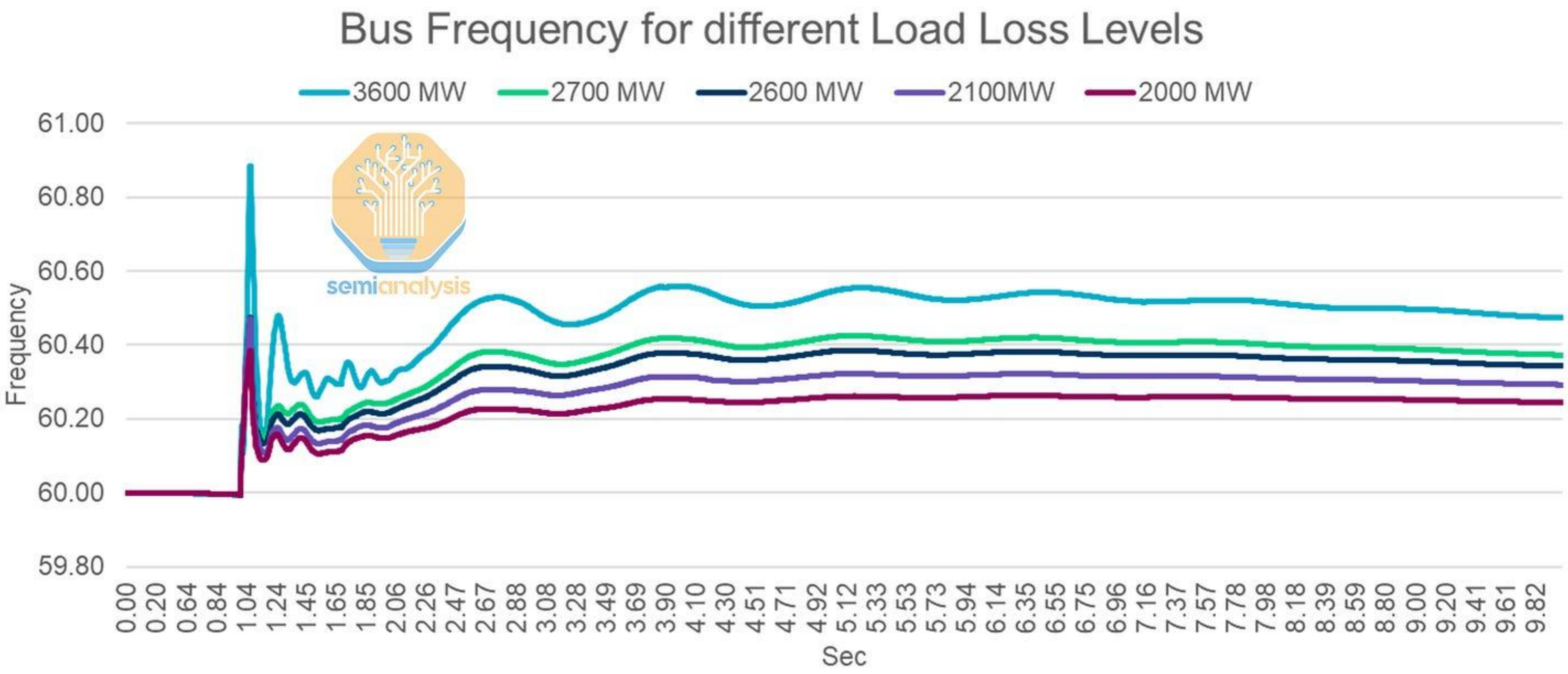
Datacenter Disconnection

Base Case Data Center Disconnection Risk		
Study Cases	**Immediate Trip**	**LVRT 20 ms**
Summer Peak (SP)	~1,500 MW	~1,500 MW
High Renewable Minimum Load (HRML)	~2,500 MW	~1,500 MW

Data Center Disconnection Risk + SynCon		
Study Cases	**Immediate Trip**	**LVRT 20 ms**
Summer Peak (SP)	~1,500 MW	~1,500 MW
High Renewable Minimum Load (HRML)	~2,500 MW	~1,500 MW

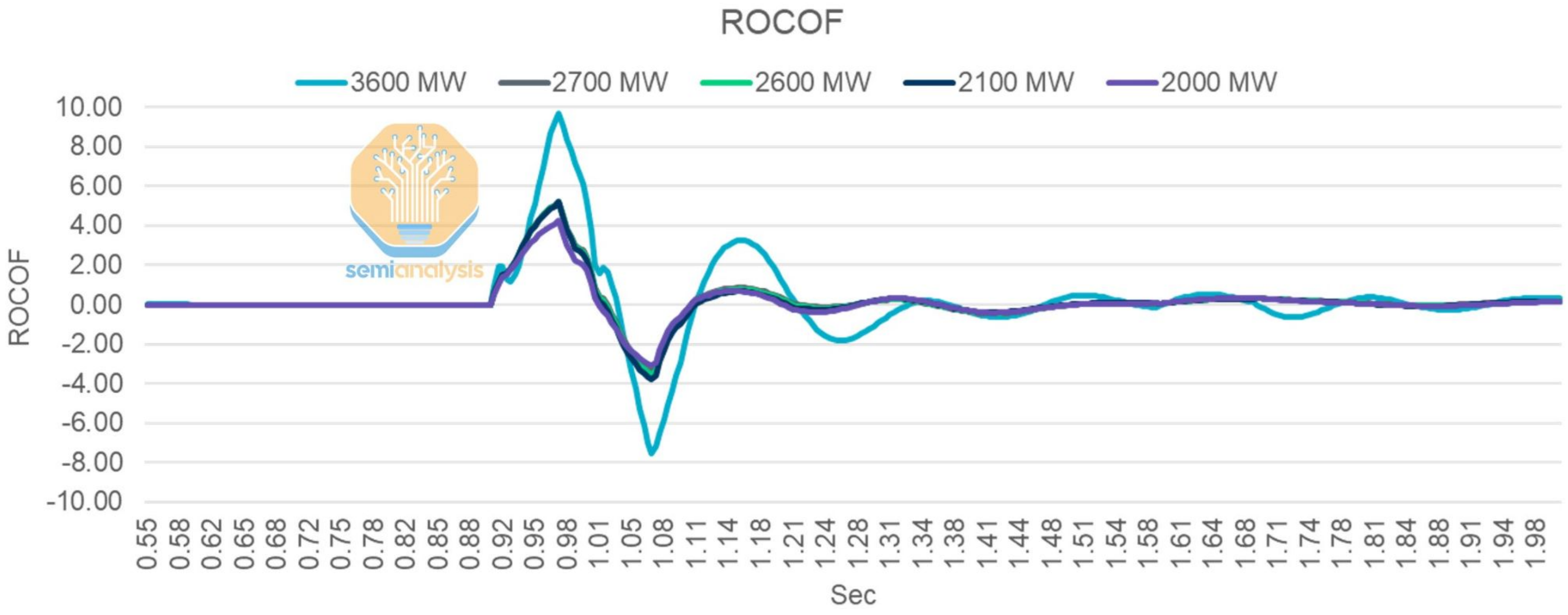
Cascade Risk Case Study

What happens to the grid *after* 2–2.5 GW of datacenters disconnect?



Cascade Risk Case Study

High value of RoCoF disables the systems ability to adjust



MW Level	Max Absolute ROCOF
~3600	**9.70**
~2700	**5.23**
~2600	**5.21**
~2100	**5.18**
~2000	4.25

Scenario (Load Loss Level - MW)	Status
~3,600	Insecure - Wide Area Voltage Issues
~2,600	Insecure - Wide Area Voltage Issues
~2,450	Insecure - Wide Area Voltage Issues
~2,100	Insecure - Local Voltage Issues
~2,000	Insecure - Local Voltage Issues
~1,950	Secure

Iberian Peninsula

Iberian 2025 Blackout: 2.2 GW generation tripped

- Cascading voltage/frequency swings: full grid collapse in 27 s.

- Miscalculated dispatch + weak interconnection: unable to stabilize.

Texas Parallel: 2–2.5 GW datacenter load drop could trigger the same cascade.

- Texas has only 4 external ties

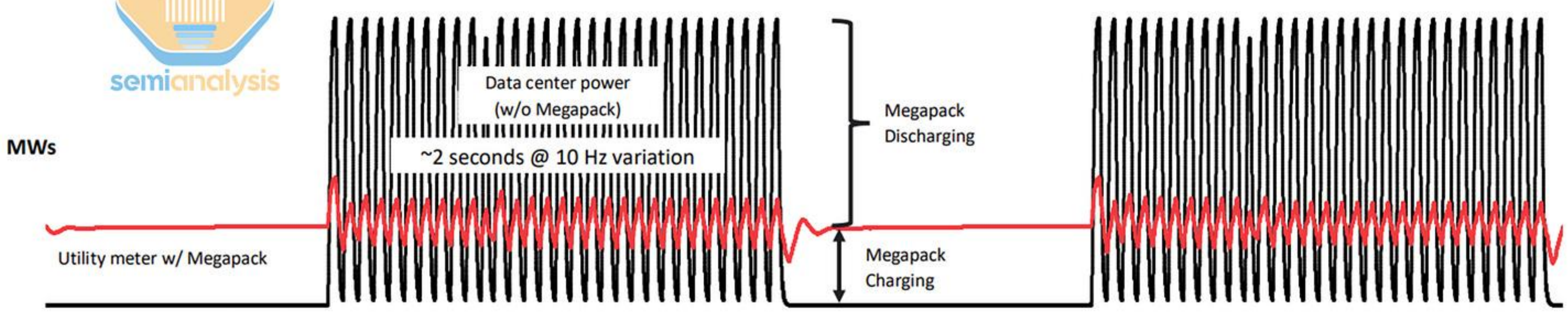
Solution: Bess

Battery Energy Storage Systems

Bess Power Quality and Stability

Megapack

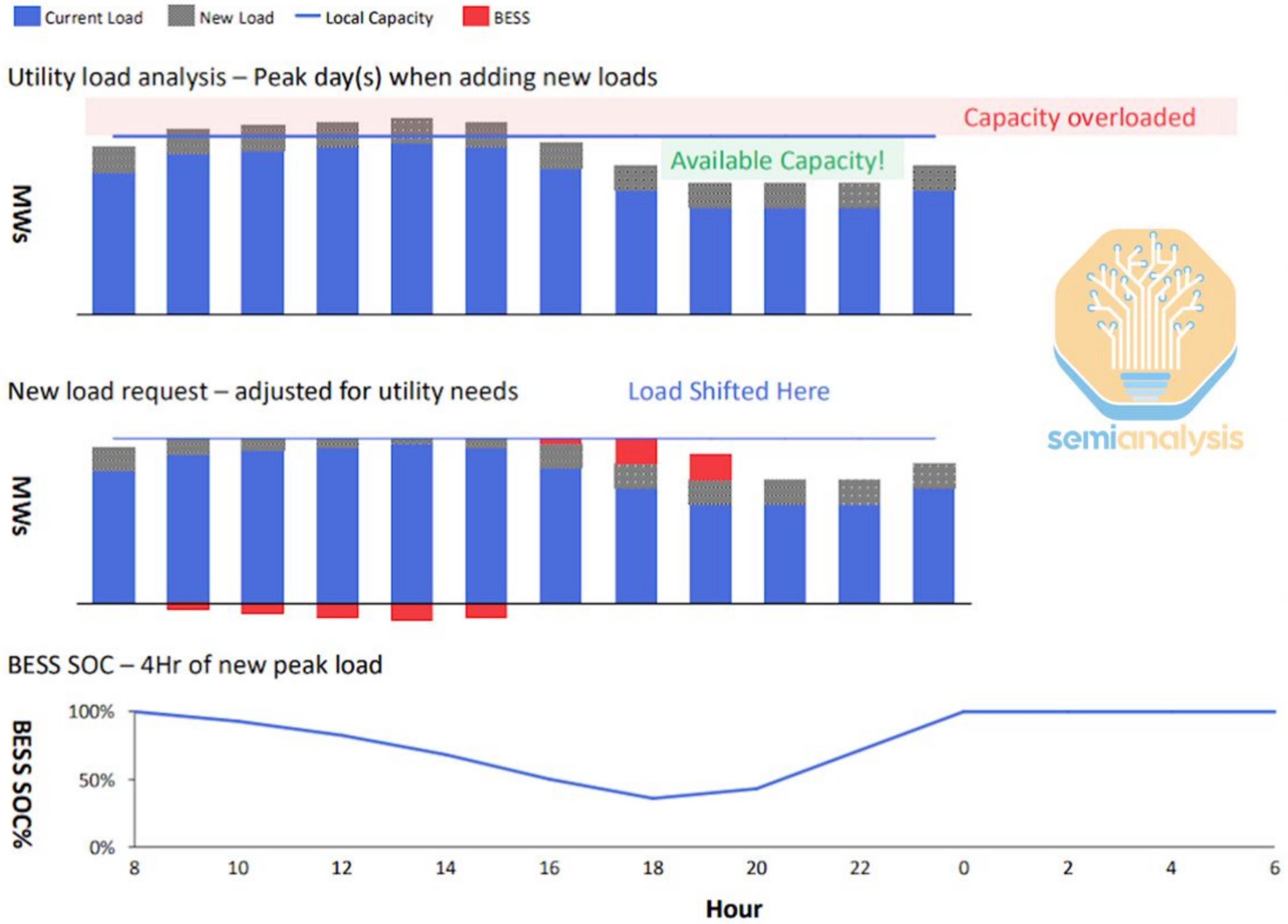
Solution: Load Smoothing with Megapack can reduce 70%+ of variability



Connecting Megapack in parallel to the load helps reduce variability → Improves grid reliability & power quality

- Energy throughput modeling shows 20+ year lifetime
- Charging and discharging are balanced such that BESS SOC is maintained for a 24/7 smoothing operation

Bess Demand Response



Load As-Is

- Can not support w/ existing infrastructure
- Likely delays in energizing new load

Adjusted Load (with BESS)

- Can support w/ existing infrastructure
- Shaping possible w/ 4hr BESS

Bess Costs

Lazard June 2024 LCOE report:

- a 100 MW BESS would cost $38-80M for a two-hour battery
- $76-157M for a four-hour battery
- Adds construction time, land use, supply-chain risks, and operational complexity
- Tesla does not consider BESS a replacement for a UPS or a diesel generator

Power Stabilization for AI Training Datacenters

Microsoft, OpenAI, NVIDIA (2025)

Background: GPU Power

Thermal Design Power (TDP): maximum power a GPU can generate in normal operation

- Maintained at granularity of 1 second

Electrical Design Power peaks (EDPp): power spikes above the TDP that GPUs can generate at small timescales

- ~50 ms timescales

Uninterruptible Power Supply (UPS): external source of overall power to a datacenter from the grid

Power Distribution Unit (PDU): device that distributes power from the UPS to individual devices in a datacenter

Background: Utility Specifications

Time-Domain Specs: constraints on *how quickly* power draw can change

- **Ramp Up/Ramp Down Rate**: maximum allowed increase/decrease in power over a given interval (MW/s)
- **Dynamic Power Range**: maximum allowed instantaneous fluctuation in power draw over a given interval

Frequency-Domain Specs: constraints on *what frequencies* emitted power may oscillate at

GPU Power Draw Profile (Hz)

Nearly all GPU energy draw occurs at frequencies < 3 Hz

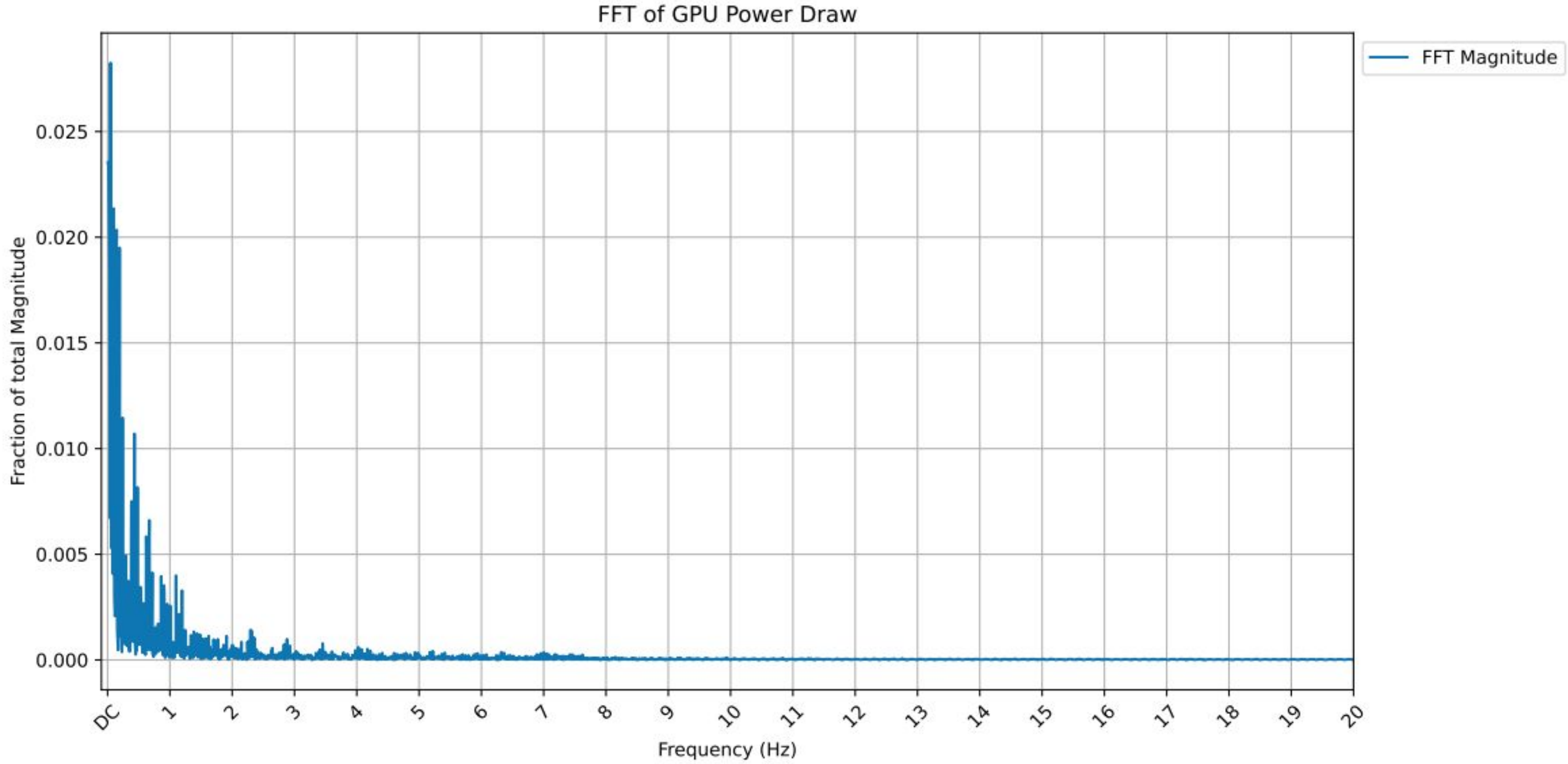


Fig. 3. Frequency components of the power waveform shown in Figure 1.

Goals

Mechanism(s) to **mitigate large, synchronous power swings**.

Must be:

- **Flexible** – Specifications differ by region and provider
- **Performant** – Increasing latency/inserting stalls undermines model development
- **Energy Efficient** – Increasing net energy consumption harms cost-effectiveness & sustainability
- **Stable** – Allowing frequent EDP peaks may violate utility spec

Software Mitigation

Introduce Secondary Workloads to Mask Dips

Secondary Workloads

Dynamically introduce **high-power workloads** when GPU power falls below pre-specified threshold.

- Either a useful job or an artificial task
  - Useful job requires saving & restoring state, which impacts primary workload performance
  - Artificial task wastes energy
- Authors built **Firefly**: increases power draw up to 100% of TDP
  - Used GPU block monitors to track activity
  - Matrix Multiplications used as secondary workload (scaled to meet need)

Challenges

- Secondary workload's ramp up/ramp down procedure harms performance of primary workload
    - Achieved < 5% performance overhead, but required lots of work on CPUs
- Difficult to deploy in cloud environments
- Shared Fate: If GPUs fail, the power management also fails
- Artificial secondary workload wastes energy

Future Optimizations

- Separate failure domains
- Predict upcoming communication phases
  - Reduce performance impact from secondary ramp up/ramp down
- Develop priority scheduling mechanisms
- Deploy software solution directly in firmware

Feasibility

Pros

- Readily-deployable
- Cheap
- Flexible
- No specialized hardware

Cons

- Either wastes energy or harms performance
- Shares fate with primary workload
- Tricky to deploy in cloud environments

Hardware Mitigation

Utilize GPU Power Smoothing to Control Power Draw

GPU Power Smoothing

Some GPUs support **programmed profiles** specifying:

- **Ramp up/ramp down** rate
- **Minimum Power Floor** (MPF): minimum power used during stable execution periods
- **Stop Delay**: duration to maintain MPF without activity before ramping down

Benefits from precision of GPU's power controller.

GPU Power Smoothing

GPU ensures power draw stays above minimum threshold

Fig. 6. Power smoothing to the minimum power floor (MPF) simulated on the training waveform from Figure 1.

Challenges

- Wastes Energy
  - 10.5% energy overhead in previous figure
- Increased power draw shortens lifespan of GPU power smoothing feature
- Constraints on MPF and EDP parameters
  - May be unable to meet tight Dynamic Power Range specifications

Feasibility

Pros

- Precise controls
- Lower performance impact
- Programmable
- Built directly into hardware

Cons

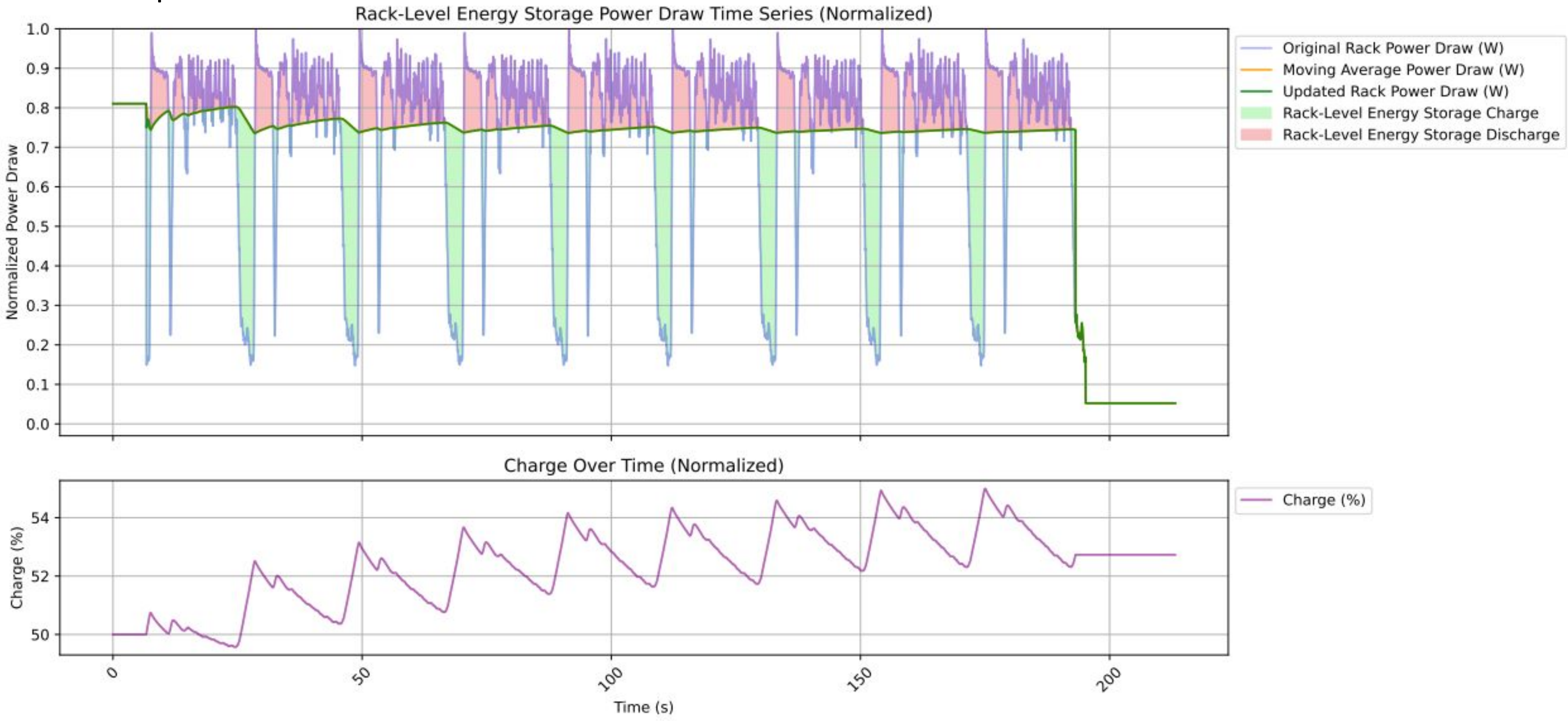
- Wastes energy
- Limited durability
- Challenge meeting tight utility specs

Energy Storage

Introduce an Energy Storage Device that Loads Power During Dips

Energy Storage

Deploy **energy storage units** in data centers that charge during communication & discharge during computation.



Requirements

- **Directly measure** power load
- Meet **sudden rises/drops** in power
- Sufficiently **high capacitance** to support training workloads
- **Quickly switch** between charge & discharge

Challenges

- Large-scale, data center-wide system requires space
    - Prefer rack-level storage devices
- Training's lower frequencies are more challenging to filter
- Ramp up/ramp down require <u>high</u> capacitance
    - Expensive
    - Bulky
    - High up-front carbon expenditure

Feasibility

Pros

- Flexible
- Meets tight specs
- Reduces overall power use
- No effect on primary workload
- Durable

Cons

- Expensive
- Placement challenges
- High initial carbon footprint

Hybrid

Manage Trade-offs with a Combination of Solutions

Summary of Solutions

No single solution is perfect.

Solution	Reliability	Performance	Energy	Cost	Ability to Meet Tightest Spec	Dependency on Developer	Lifetime
Software	Medium	Medium	High	Medium	High	High	High
GPU	High	Medium	High	Low	Medium	Medium	Medium
Storage	High	High	Low	High	High	Low	High

*Green = good, red = bad

Combination Approach

Authors' Proposal: Employ GPU power smoothing, add software loads during down periods, and deploy rack-level energy storage.

- GPU addresses ramp up/ramp down specs
- Injecting software loads prevents intermediate fluctuations
- GPU & storage would have to communicate info on level of charge
- Larger future projects could use larger battery storage

Future Work

Systems Opportunities:

- **Asynchronous training** techniques
- Novel **scheduling** approaches
- **Overlap** of compute & communication
- **Power-aware** training algorithms

Collaboration is Required:

- Create **open forums** to develop shared standards for telemetry, oscillation mitigation, etc.
- Standardize utility specs & establish **dedicated communication** pipelines with datacenter operators